

Pauline Emma Michèle De Baets

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PROFILE

EPFL Microengineering student with solid experience in embedded systems, microcontrollers, and hands-on hardware projects. Focused on robotics and autonomous systems, with practical experience designing, manufacturing, and testing rocket recovery mechanisms.

EDUCATION

EPFL – BSc in Microengineering (2023–2026, expected) GPA: 4.9/6
Third-year exchange at TU Delft (Electrical Engineering Faculty) (2025–2026).
Collège Sismondi, Geneva – Bilingual Maturité (2019–2023) GPA: 5.8/6
Specialisation in physics and applied mathematics.

TECHNICAL SKILLS & OTHER

- **Programming languages:** C++, C, Python, AVR assembly.
- **Embedded & Electronics:** Embedded systems (ATmega128L), microcontrollers, basic digital/analog electronics.
- **Mechanical & CAD:** 3D/2D CAD (SolidWorks, CATIA), mechanical design, 3D printing.
- **Other:** Introductory Machine learning (coursework and projects, including radar-based gesture recognition and pedestrian detection using Python and Numpy).
- **Interests:** Football, tennis, alpine skiing, and piano.

LANGUAGES

French: Native • **English:** C2 (Cambridge Proficiency) • **German:** B2 • **Dutch:** Intermediate

EXTRACURRICULAR

- **EPFL Rocket Team - Recovery Division** (2024-2025)
Responsible for the design, manufacturing and testing of a line cutter system, used to initiate the second descent phase by severing the parachute's reefing line.
- **Robopoly - EPFL Robotics Association** (2023-2025)
Assembled a Prism robot kit and gained 3D-printing and basic electronics prototyping experience; currently co-developing a small car project with peers.

SELECTED ENGINEERING PROJECTS

- **Line cutter (Rocket Team):** Designed, manufactured, iterated and tested a compact pyrotechnic device using an e-match-triggered charge and a guided blade to cut high-strength Dyneema reefing lines, triggering the second phase of the rocket's descent.
- **Force Sensor - Mechanism Conception Group Project (EPFL):** design of a 3-DOF force sensor capable of measuring contact forces in all directions with high precision, using compliant mechanisms and flexures, force and moment balancing and capacitive displacement sensors. Developed the full mechanical concept, modelling and dimensioning to meet industrial constraints from Hexagon.
- **Pac-Man Style Game on ATmega128L (EPFL):** Built a full Pac-Man style game on the ATmega128L entirely in AVR assembly, integrating WS2812B LED matrix graphics, a PS/2 keyboard user interface, LCD display, and an internal EEPROM profile storage system. Implemented a multi-level maze, timer-controlled enemy movement, and collision detection.
- **Radar-based hand gesture recognition (TU Delft):** Developed a machine learning pipeline for automatic classification of four hand gestures (wave, pinch, swipe, click) from micro-Doppler radar spectrograms (DopNet dataset), including preprocessing, feature extraction, and a neural network evaluated on an unseen test set.

Relevant project grades: Force Sensor 6/6, Pac-Man 5.5/6, Radar-based hand gesture recognition 8/10

EXPERIENCE

- **Volunteer - Ilanga** (April - May 2022, 1.5 months)
Taught daily English classes to children and teenagers (ages 3–18) in a community programme supporting destitute children and mothers.
- **Internship - Fiduciaire de Preux** (May 2022, 2 weeks)
Introduction to the operational, legal, and digital aspects of financial management. Contributed to daily administrative and organizational processes, and gained insight on document and data management, ISO-certified processes as well as professional communication and client interface.
- **Internship - Creolab Orthodontics Laboratory** (June - July 2022, 1 month)
Worked on digital workflows involving 3D scanning of dental models, precise segmentation of the models, and high precision resin printing for orthodontics applications.